

**NEXT-GENERATION DENTAL AI:
DEEP LEARNING-BASED RADIOGRAPHIC
ANALYSIS WITH MULTIMODAL CLINICAL
INTELLIGENCE**

A PROJECT REPORT

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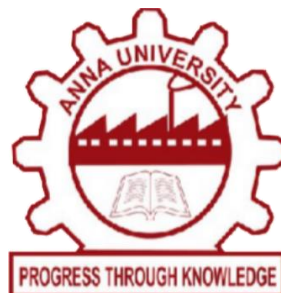
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BONAFIDE CERTIFICATE

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DECLARATION

We affirm that the project work entitled "**NEXT-GENERATION DENTAL AI: DEEP LEARNING-BASED RADIOGRAPHIC ANALYSIS WITH MULTIMODAL CLINICAL INTELLIGENCE**" being submitted in partial fulfilment for the award of the degree of **BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND ENGINEERING** is the record of our own work carried out under the guidance of **Mrs. R.B.RAMA DEVI, M.E., Assistant Professor, Department of Computer Science and Engineering, R.V.S. College of Engineering, Dindigul**. This report has not been submitted for the award of any other degree, diploma, fellowship, or other similar titles in any other University or Institution.

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ABSTRACT

The rapid proliferation of digital dental imaging, particularly **panoramic radiographs** (Orthopantomograms, OPGs), has created a critical need for intelligent systems capable of assisting clinicians in accurate and efficient diagnosis. Manual interpretation of these radiographs is cognitively demanding, time-intensive, and subject to significant inter-examiner variability, often reported between 15% and 35%. This project addresses these challenges through the design and implementation of an advanced AI-driven platform titled **“Multimodal Dental Diagnostic and Clinical Reporting System” (DentalAI)**, which harmonizes deep learning-based visual detection with multimodal clinical reasoning.

The core of the system utilizes a **YOLOv8x-segmentation** architecture for simultaneous object detection and pixel-level mask generation of dental anomalies. The model was fine-tuned using transfer learning on a dataset comprising **31 dental classes**, including pathologies (caries, lesions), restorations (fillings, crowns), and orthodontic appliances. Training was conducted in a high-performance environment using an **NVIDIA Tesla T4 GPU**, demonstrating stable convergence with **box loss** decreasing to 1.4585 and **classification loss** reaching 0.9993. The model achieves a box detection **mAP@50 of 0.3059** and a segmentation **mAP@50 of 0.2842**. Notably, precision-confidence analysis

reveals a **Precision of ~0.87** at high-confidence thresholds, ensuring reliable predictions. The system operates with an impressive inference latency of **12–15 milliseconds**, making it suitable for real-time clinical deployment.

A key innovation of this research is the integration of **LLM MODEL (Google Gemini 2.0)** for multimodal clinical reasoning. The system synthesizes three distinct data streams: **radiographic image input**, AI-generated detection coordinates, and clinician voice notes processed radiographic image input via a **Voice-to-Clinical Record** feature. To ensure production-grade stability, a **"Resilience Architecture"** was implemented, featuring automated API key rotation and a multi-model fallback strategy. The final output is an **FDI-compliant structured clinical report**, including annotated overlays and treatment recommendations in PDF format. Experimental results indicate that the system can reduce documentation time by approximately **93%**, significantly optimizing diagnostic throughput and standardizing dental reporting.

Keywords: *YOLOv8-Segmentation, Gemini 2.0 Flash, OPG, Multimodal AI, FDI Notation, Clinical Reporting, Dental Informatics, Resilience Architecture.*

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation / Symbol	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
AMP	Automatic Mixed Precision
ASGI	Asynchronous Server Gateway Interface
BCE	Binary Cross-Entropy
CBCT	Cone Beam Computed Tomography
CAD	Computer-Aided Diagnosis
CDN	Content Delivery Network
CI/CD	Continuous Integration / Continuous Deployment
CIoU	Complete Intersection over Union
CNN	Convolutional Neural Network
CORS	Cross-Origin Resource Sharing
CSP	Cross-Stage Partial
C2f	Cross Stage Partial bottleneck with 2 fused convolutions
DFL	Distribution Focal Loss
DFD	Data Flow Diagram
EHR	Electronic Health Record
FDI	Fédération Dentaire Internationale (World Dental Federation)
FPN	Feature Pyramid Network
GFLOPs	Giga Floating Point Operations Per Second
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
HIPAA	Health Insurance Portability and Accountability Act
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IoU	Intersection over Union

JSON	JavaScript Object Notation
LLM	Large Language Model
mAP	Mean Average Precision
mAP@50	Mean Average Precision at IoU threshold 0.50
mAP@50-95	Mean Average Precision at IoU thresholds 0.50 to 0.95
MUI	Material User Interface
MIME	Multipurpose Internet Mail Extensions
NLP	Natural Language Processing
NMS	Non-Maximum Suppression
OPG	Orthopantomogram (Panoramic Dental Radiograph)
ONNX	Open Neural Network Exchange
PAN	Path Aggregation Network
PaaS	Platform as a Service
PDF	Portable Document Format
PNG	Portable Network Graphics
REST	Representational State Transfer
RCT	Root Canal Treatment
SPA	Single Page Application
SPPF	Spatial Pyramid Pooling Fast
SSD	Single Shot Detector
STT	Speech-to-Text
SVG	Scalable Vector Graphics
T4	Tesla T4 (NVIDIA GPU model used for training)
UI	User Interface
URL	Uniform Resource Locator
VGG	Visual Geometry Group (CNN architecture)
ViT	Vision Transformer
VLM	Vision Language Model
VRAM	Video Random Access Memory
YAML	Yet Another Markup Language
YOLO	You Only Look Once
α	Alpha — trade-off parameter in CIoU loss
ρ	Rho — Euclidean distance between box centres in CIoU
σ	Sigmoid activation function
lr0	Initial learning rate

lrf	Final learning rate ratio
L_CIoU	Complete IoU bounding box regression loss
L_cls	Classification loss
L_DFL	Distribution Focal Loss
L_mask_bce	Mask Binary Cross-Entropy segmentation loss
L_mask_iou	Mask IoU segmentation loss
L_total	Total combined training loss